NATIONAL RESEARCH UNIVERSITY
HIGHER SCHOOL OF ECONOMICS

Faculty of Computer Science
Bachelor's Programme "Data Science and Business Analytics"

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Research Project Report on the Topic:

Searching for patterns in simplices

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Object of research: "Searching for patterns in simplices"

It should be clear that simplices are the subject of the research in the "Search for Patterns in Simplices" project. Nevertheless, we must begin by comprehending what pattern search is, then separately take into account what simplices are. Following all of this, it will be easy to comprehend pattern search in simplices precisely and its operation.

The relevance of research:

The search for patterns in simplices is the topic of this project's study. However, in general, the study of patterns in simplices might be essential for various professions, such as mathematics, computer science, data science, etc. The applicability of this project relies on the particular subject in which it is used.

Introduction about the subject of research:

Of course, a different theory that will aid in the study of simplices and patterns will be the research's subject. As I've already managed to realize, the majority of the research will come to fruition not only through the understanding of a few mathematical and geometrical principles, but also through the reading of numerous articles that will, in the end, help me gain good knowledge in this field and produce an effective research outcome.

Work plan:

1. Simplices
2. Simplicial complexes
3. Graph isomorphism
4. Whitney's theorem
5. Graph homeomorphism
6. Barycentric division
7. Random graph
8. Further research (producing the code)

Subject of research:

Simplex

Simplex is the simplest convex polyhedron of a given number of dimensions n:
* n = 3: a three—dimensional simplex is an arbitrary, including an irregular, tetrahedron;
* n = 2: a two-dimensional simplex is an arbitrary triangle;
* n = 1: under one—dimensional - segment;
* n = 0: a zero-dimensional simplex is just one point.

n the dimensional simplex has $n+1$ a vertex that collectively does not belong to any $(n-1)$ the dimensional subspace of the Euclidean space (with the number of dimensions n or more) in which this simplex lies. Any vertices, $r+1$ vertexes, $0 \leq r \leq n-1$, taken from the data number $n+1$ vertexes n - dimensional simplex, define some r - dimensional simplex — r - dimensional face of this simplex. The zero-dimensional faces of a simplex are its vertices, the one-dimensional faces are called edges.

Simplicial complexes

Definition 1.1. A simplicial complex on a finite set of vertices M is called a set $K \subset 2^M$
subsets of the set M satisfying the following two conditions:
1. if
$I \in K$ и $J \subset I$, то $J \in K$;
2.
$\varnothing \in K$.

The elements of the set M are called vertices of the simplicial complex K, the elements I ∈ K are its simplices, and if i ∈ I, then we say that i is the vertex of the simplex I. The set of vertices we will sometimes denote Vert(K). The number |I| - 1 is called the dimension of the simplex I and is denoted by dim I. The dimension of a simplicial complex K is by definition the maximum dimension of its simplices. Note that the formally empty simplex has dimension -1.

Note that it does not follow from the definition that a vertex (considered as a singleton set) is a simplex. Let's call a vertex i ∈ Vert(K) ghostly if {t} ∉ K. A ghost vertex cannot be a vertex of any simplex in K, as the definition easily implies. Therefore, as a rule, ghostly peaks can be thrown out without much harm. In the future, it will be assumed that the simplicial complex does not have ghost vertices, and in those rare cases when ghost vertices do arise and it does not work out to get rid of them, or it does not make sense, this will be specifically stipulated.
We used to think of graphs in terms of pictures. It is also useful to be able to turn an abstract simplicial complex into a picture (i.e. into a topological space). This topological space (called the geometric realization of a simplicial complex) can be described in the same way as it is with a graph: a point ei is assigned to each vertex i P M, a segment between ei and ej is drawn for each edge {i, j} ∈ K, for each triple{i, j, k} ∈ K draws a triangle stretched over ei, ej, ek, and so on. Also, if we draw it in some space, then we want this unit not to self-intersect where it should not. To do this, the points ei must be taken in a sufficiently general position in a space of sufficiently large dimension, which is the motivation for the following definition.

Definition 1.2. Let K be a simplicial complex on the rms set. Let e1,...,em be the basis of Rm space, and let $\triangle_I = \mathrm{conv}(e_i \mid i \in I)$ is a simplex spanned by the basis vectors corresponding to the indices from the subset $I \subset [m]$. A subset $R(K) = \bigcup_{I \in K} \triangle_I \subset \mathbb{R}^m$ of is called the standard implementation of a simplicial complex K. Induced with $\mathbb{R}^m$ topology defines on R(K) the structure of a compact topological space. This topological space is denoted by |K| and is called the geometric realization of the simplicial complex K.

Definition 1.3. A simplex I ∈ K is called maximal by inclusion if there is no such J ∈ K that J strictly contains I. A simplicial complex K is called pure (or dimensionally homogeneous) if all its maximal inclusion simplices have the same dimension.

Definition 1.4. Simplicial complexes K and L are called isomorphic if there is a bijection between the sets of their vertices inducing a bijection between K and L.

Definition 1.5. Let K1 and K2 be simplicial complexes on sets
M1 and M2 respectively. A joint K1*K2 is a simplicial complex on a 5 disconnected union $M_1 \sqcup M_2$, whose simplices have the form $I_1 \sqcup I_2 \subset M_1 \sqcup M_2$, where $I_1 \in K_1$, $I_2 \in K_2$.

Definition 1.6. Let $J \subset M$ is an arbitrary subset of vertices. A simplicial complex $K_J \stackrel{\text{def}}{=} \{I \in K \mid I \subseteq J\}$ is called a complete subcomplex of a complex K on the set J.

Definition 1.7. Let I ∈ K. The star of simplex I in complex K is called the def simplicial complex $\operatorname{star}_K I \stackrel{\text{def}}{=} \{J \subseteq M \mid J \cup I \in K\}$. The link of a simplex I in a complex K is called a simplicial complex $\operatorname{link}_K I \stackrel{\text{def}}{=} \{J \subset M \backslash I \mid J \sqcup I \in K\}$.

Graph isomorphism

Graph theory says that graph isomorphism occurs when between two elements: $G = \langle V_G, E_G \rangle$ and $H = \langle V_H, E_H \rangle$. Between the sets of vertices in these graphs, there is a bijection: $f\colon V_G \to V_H$
As long as vertices f(u) and f(v) are contiguous in graph H, any two vertices u and v from graph G will be close together. As a starting point, undirected graphs without vertices or edges assigned weights are used. The "orientation of arcs and the values of weights" must be preserved under extra conditions if the isomorphism notions are to be applied to an oriented or weighted graph. The symbol for the graph isomorphism is $G \simeq H$.

For example, you can pay attention to two isomorphic graphs and what conditions are met for this:

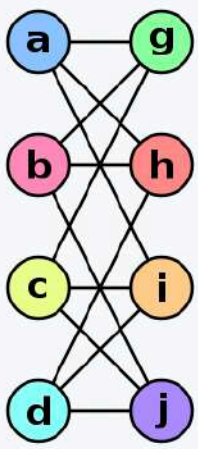
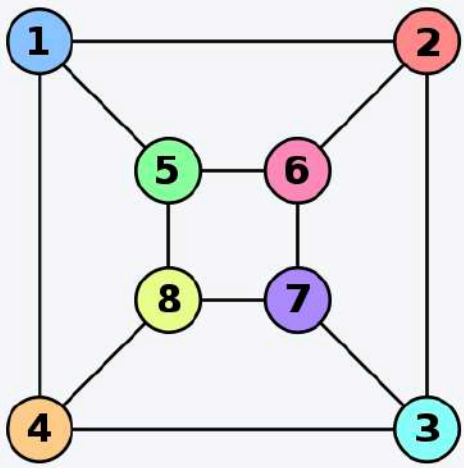
Graph G	Graph H	An isomorphism between G and H
		$f(a) = 1$ $f(b) = 6$ $f(c) = 8$ $f(d) = 3$ $f(g) = 5$ $f(h) = 2$ $f(i) = 4$ $f(j) = 7$

Figure 1. Two isomorphic graphs

Whitney 's Theorem

Hassler Whitney in 1932 developed this graph isomorphism theorem. According to the theorem, two graphs are only isomorphic if their "edge graphs" are also isomorphic. A complete 3-vertex graph and a complete 2-lobed graph are the exceptions. The bipartite graph is shown on the right and the exclusion graph with three vertices is shown in the image below. Although their edges are isomorphic, the graphs themselves are distinct, as can be seen.

Figure 2. Two bipartite graphs.

Graph homeomorphism

Only when two subgroups of two distinct graphs exhibit isomorphism can two graphs be said to be homeomorphic. Two graphs are considered homeomorphic in the context of graph theory when they are homeomorphic in the topological sense. This is based on the assumption that a graph's edges are the segments that link its vertices.

Vertices can also be an exception, but first, let's define the term graph subdivision to mean a graph created by splitting the edges of another graph. To illustrate, consider a graph with vertices u, v and an edge e. This subdivision creates two edges, each with a vertex w between u and v.

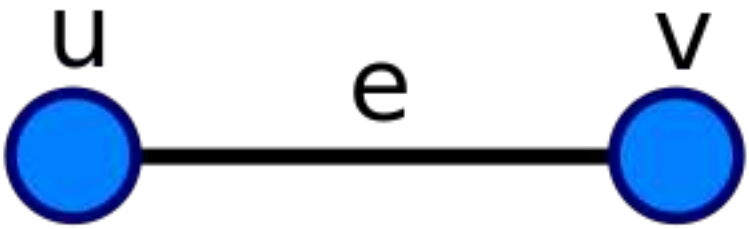


Figure 3. Graph with two vertices
.

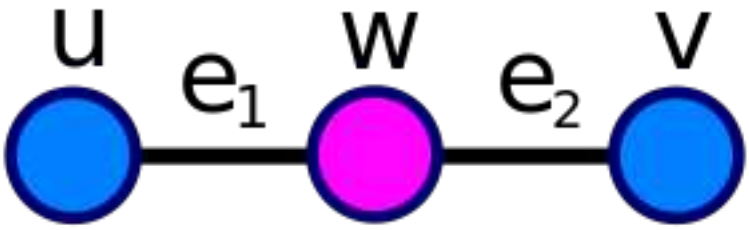


Figure 4. Graph with three vertices.

The reverse subdivision procedure is the exception. We may create a network with two vertices by removing vertex w and linking vertices u and v when we have two edges (e1 and e2) with the vertices u, w, and v. It's also important to remember that this technique only works on vertices that are "incent" to precisely two edges.

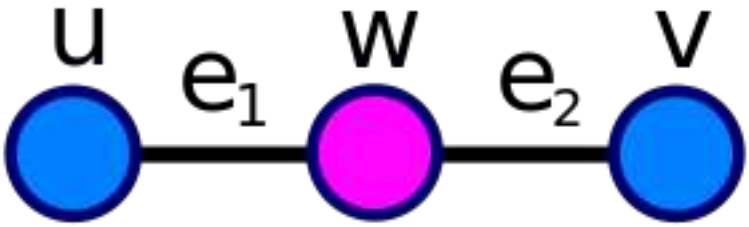


Figure 5. Graph with three vertices.

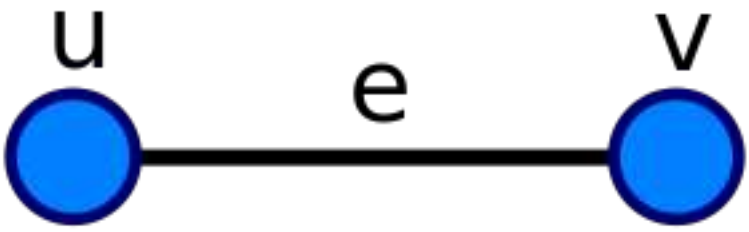


Figure 6. Graph with two vertices.

Barycentric divisions

Each edge of the graph is divided by this subdivision, producing a bipartite graph in every case. It is possible to carry out this process repeatedly to achieve the condition where the n-th barycentric subdivision is the barycentric subdivision of the n-1 subdivision. Subdivision is repeated until a simple graph is formed.

The term bipartite graph or Bigraph refers to a graph whose set of vertices can be split into two parts, with each edge connecting vertices from distinct portions of the graph and no edges connecting vertices from the same part of the graph.

Random graph

In actuality, this is the graph's probability distribution, indicating that "random graphs" are best described by a random process that creates them. Graph theory and probability theory are combined in this topic. Random graphs are a crucial tool for the study of common graph features from a mathematical perspective. There are several random graph models that capture the intricate network patterns found in different sectors. These models are frequently employed in fields where complicated network modeling is necessary. A «random graph» is produced by systematically linking random isolated vertices from a set of vertices. The basic goal of random graph modeling is to pinpoint the precise instant at which certain network features emerge. A number of probability distributions are generated on the graph using various models of random graphs. The Hilbert distribution G(n,p), in which each potential edge arises independently with probability 0 p 1, is one of the most researched models of random graphs. This implies that there is a chance that each pair of vertices will be joined by an edge. Based on this approach, it is possible to determine the likelihood of finding a graph with a particular number of edges: $\binom{N}{m} p^m (1-p)^{N-m}$.

The Erdos-Renyi model G(n,M) is another model of random graphs, where any graph with n vertices has the same probability of having precisely M edges. This implies that we fix the edge count and take into account all feasible graphs with that edge count. $N = \binom{n}{2}$

The model G(n,p) may be viewed as a random process that begins with n isolated vertices on a graph and gradually adds edges. With the same probability, one edge is chosen from a group of still unconnected vertices at each phase $\binom{N}{M}^{-1}$. We obtain an infinite random network if we take into account the limiting scenario with an infinite number of vertices and separately choose each potential edge with probability 0 p 1. This model establishes the likelihood of each edge's existence, and it may be investigated using probabilistic techniques.
Thus, we may explore the statistical features and traits of graphs of different shapes and dimensions using random graph models.

Further research:

The next task of my project was the need to write code for simplices, namely, it was necessary to learn how to generate random simplicial complexes of a certain size, where the parameter was vertices that could be changed. The next step is to select some small simplicial complex and investigate how many of these small complexes were found inside a large simplicial complex. This experiment had to be repeated at least 1000 times and see how many such subcomplexes turned out. Next, it was necessary to calculate the mean and standard deviation, and see how they change depending on the vertices. As a basis, I took 10 vertices and the number of repetitions equal to 1000.

So I took the 1-dimensional simplex as a basis:

Size	Mean	Std Dev
2	1.0	0.0
3	3.0	0.0
4	6.0	0.0
5	10.0	0.0
6	15.0	0.0
7	21.0	0.0
8	28.0	0.0
9	36.0	0.0
10	45.0	0.0

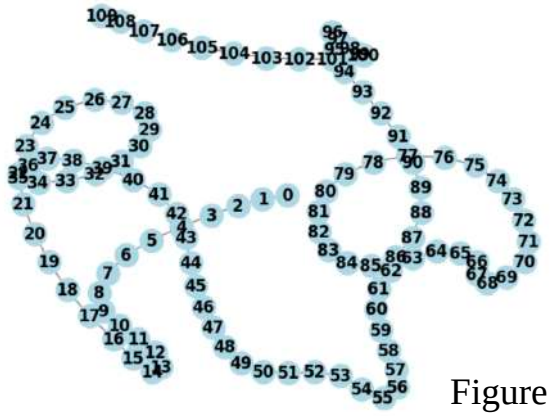


Figure 7. 1-dimentional simplex.

Which suggests that the number of subcomplexes increases with the size of the simplex. For example, with a simplex size of 2 (segment), there is 1 subcomplex of size 2 (segment itself). With a simplex size of 3 (triangle), there are already 3 subcomplexes of size 2 (3 segments). With a simplex size of 4 (tetrahedron), the number of subcomplexes of size 2 increases to 6 (6 segments), and so on.
The standard deviation (Std Dev) for all simplex sizes is 0, which means that all values are the same and there is no spread in the data.
These results show a pattern in the number of subcomplexes of their simplices of various sizes.

For a 2-dimensional cyclic simplex, I got:

Size	Mean	Std Dev
2	0.49	0.4998999899979995
3	1.899	1.1527354423283775
4	4.76	1.8100828710310475
5	8.814	2.1774765211133733
6	14.094	2.1538718624839315
7	20.31	2.072172772719495
8	27.611	1.6750161193254234
9	35.752	1.3865410199485626
10	44.838	1.1965600695326586

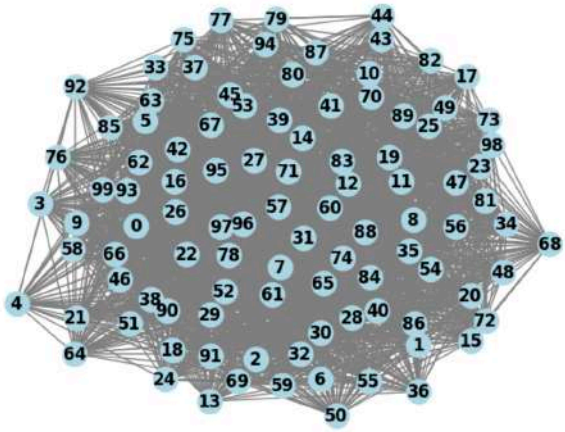


Figure 8. 2-dimentional cyclic simplex.

This shows that the number of subcomplexes also increases with the size of the simplex. However, a more complex relationship is observed, in contrast to the 1-dimensional simplex, where a linear increase was observed.

It can be noticed that with increasing dimensionality of the simplex, the average number of small simplicial complexes found also increases. This indicates that when new vertices are added to the complex, more possible combinations appear to form small complexes. Also, the standard deviation increases with increasing dimensionality of the simplex. Which means that the variability of the found complexes becomes larger with increasing dimensionality.

For a 2-dimensional acyclic simplex, I got:

Size	Mean	Std Dev
2	0.506	0.4999639987039066
3	1.629	0.7066533803782447
4	3.528	0.7260964123310346
5	6.39	0.6556675987114203
6	10.226	0.5068767108479142
7	15.134	0.3768872510446592
8	21.068	0.2633932421304693
9	28.03	0.17635192088548401
10	36.016	0.12547509713086497

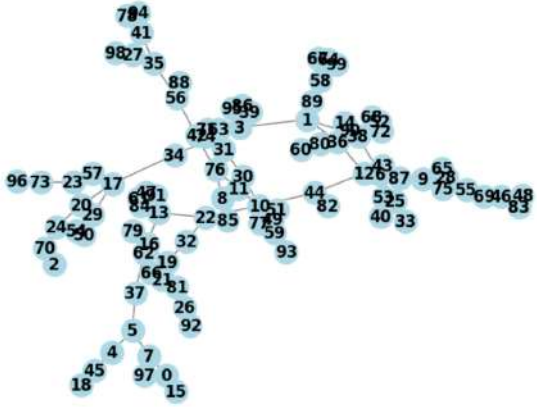


Figure 9. 2-dimentional acyclic simplex.

For the average number of small simplicial complexes found, we observe the same pattern as in the cyclic one. But the values of the standard deviation decrease with increasing dimensionality of the simplex. Which indicates that the variability of the number of small complexes found decreases with increasing dimensionality.

For a 3-dimensional cyclic simplex, I got:

Size	Mean	Std Dev
2	0.518	0.4996758949559204
3	1.63	0.7007852738178793
4	3.566	0.7480935770343173
5	6.342	0.6124018288672887
6	10.224	0.523281950768417
7	15.122	0.3783067538387334
8	21.065	0.25054939632735096
9	28.035	0.19435791725576812
10	36.025	0.16844880527923012

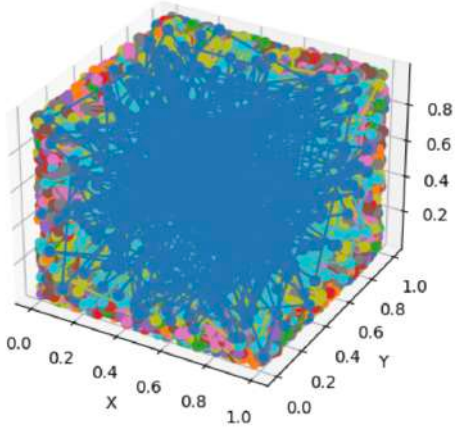


Figure 10. 3-dimentional cyclic simplex.

As in the previous cases, with an increase in the dimension of the simplex, the average number of small simplicial complexes found also increases.
But the values of the standard deviation have some variability, but in most cases they decrease with increasing dimensionality of the simplex, which indicates a more stable nature of the number of small complexes found with increasing dimensionality.

For a 3-dimensional acyclic simplex, I got:

Size	Mean	Std Dev
2	0.506	0.4999639987039066
3	1.669	0.7067099829491587
4	3.509	0.6970789051463256
5	6.41	0.6869497798238238
6	10.226	0.5068767108479142
7	15.131	0.3844983745089178
8	21.084	0.3015692291995322
9	28.039	0.19359493795035035
10	36.019	0.14366279963859815

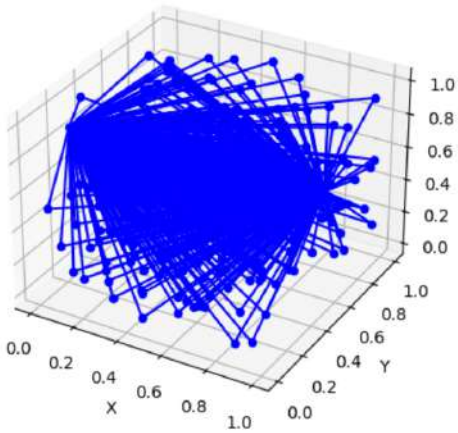


Figure 11. 3-dimentional acyclic simplex.

It can be noted that the pattern in the acyclic simplicial complex has not changed, which tells us that we observe the same pattern as before.

I also tried to write code for a multi-dimensional simplex, in which the mean and standard deviation will vary from the number of selected vertices and faces.

Work Schedule:

From 20.11.2022 to 30.12.2022 I was learning some information about simplices.

From 08.01.2023 to 21.01.2023 I was learning what simplicial complexes means and how does it work.

From 22.01.2023 to 14.02.2023 the graph isomorphism and graph homeomorphism were studied.

From 15.02.2023 to 07.06.2023 I will be practicing with code.

Result:

I managed to write the code that I planned to do by the end of the study. This code helped to draw the following conclusions: The dimension of a simplicial complex affects the number of small simplicial complexes found. With increasing dimensionality, the total number of complexes found also increases. This indicates that in higher dimensions there are more possible combinations of vertices that form small complexes. The cyclicity of the simplicial complex also affects the number of small complexes found. In cyclic complexes, the number of complexes found is usually higher compared to acyclic complexes of the same dimension. This may be due to the peculiarities of the structure of cyclic complexes, which have a large number of cycles and loops that contribute to the formation of small complexes. The standard deviation in the number of small complexes found indicates the variability of the experimental results. A lower value of std dev indicates a more predictable nature of the number of complexes found, while a higher value indicates greater variability in the results.

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